

50 Java Streams Practice Questions for SDETs

Introduction

This document contains 50 carefully designed Java Streams practice questions specifically for:

- SDETs
- Automation Test Engineers
- QA Engineers
- API Testers
- Selenium Automation Engineers

The questions are arranged in the following order:

1. Beginner Level
2. Beginner to Intermediate
3. Intermediate
4. Advanced

Each question focuses on:

- real-world automation scenarios
 - practical Stream usage
 - interview-oriented concepts
 - day-to-day SDET work
-

Level 1 — Beginner (Questions 1-15)

Concepts Covered

- stream()
 - forEach()
 - filter()
 - map()
 - count()
 - sorted()
 - distinct()
-

Question 1 — Print All Browser Names

Problem

You have a list of browser names.

Print all browser names using Streams.

Input

```
["Chrome", "Firefox", "Edge"]
```

Concept Practiced

- stream()
 - forEach()
-

Question 2 — Print All Even Numbers

Problem

From a list of numbers, print only even numbers.

Input

```
[1, 2, 3, 4, 5, 6]
```

Expected Output

```
2  
4  
6
```

Concept Practiced

- filter()
-

Question 3 — Print All Odd Numbers

Problem

Print only odd numbers from the list.

Concept Practiced

- filter()
 - lambda expressions
-

Question 4 — Convert Browser Names to Uppercase

Problem

Convert all browser names to uppercase.

Input

```
["chrome", "firefox"]
```

Expected Output

```
["CHROME", "FIREFOX"]
```

Concept Practiced

- `map()`

Question 5 — Multiply Numbers by 10

Problem

Multiply every number by 10.

Input

```
[1,2,3]
```

Expected Output

```
[10,20,30]
```

Concept Practiced

- `map()`

Question 6 — Count Total Elements

Problem

Count total number of elements in the list.

Concept Practiced

- `count()`
-

Question 7 — Count Failed Test Cases

Problem

From a list of test statuses, count how many tests failed.

Input

```
["PASS", "FAIL", "PASS", "FAIL"]
```

Expected Output

```
2
```

Concept Practiced

- `filter()`
 - `count()`
-

Question 8 — Sort Numbers in Ascending Order

Problem

Sort numbers in ascending order.

Concept Practiced

- `sorted()`
-

Question 9 — Sort Browser Names Alphabetically

Problem

Sort browser names alphabetically.

Concept Practiced

- `sorted()`
-

Question 10 — Remove Duplicate Numbers

Problem

Remove duplicate numbers.

Input

```
[1,1,2,2,3,3]
```

Expected Output

```
[1,2,3]
```

Concept Practiced

- `distinct()`
-

Question 11 — Remove Duplicate Browser Names

Problem

Remove duplicate browser names.

Concept Practiced

- `distinct()`
-

Question 12 — Print First 3 Elements

Problem

Print only first 3 elements.

Concept Practiced

- `limit()`
-

Question 13 — Skip First 2 Elements

Problem

Skip first 2 elements and print remaining.

Concept Practiced

- `skip()`
-

Question 14 — Find Names Starting with A

Problem

Print all names starting with A.

Input

```
["Amit", "John", "Alex"]
```

Expected Output

```
["Amit", "Alex"]
```

Concept Practiced

- `filter()`
 - `startsWith()`
-

Question 15 — Find Numbers Greater Than 50

Problem

Print numbers greater than 50.

Concept Practiced

- `filter()`
-

Level 2 — Beginner to Intermediate (Questions 16–30)

Concepts Covered

- `collect()`
 - `anyMatch()`
 - `allMatch()`
 - `noneMatch()`
 - `findFirst()`
 - `findAny()`
 - `reduce()`
 - `Optional`
-

Question 16 — Collect Even Numbers into List

Problem

Store all even numbers into another list.

Concept Practiced

- `collect()`
 - `Collectors.toList()`
-

Question 17 — Collect Failed Test Cases

Problem

Collect all failed test statuses into another list.

Concept Practiced

- filter()
 - collect()
-

Question 18 — Check if Any Test Failed

Problem

Check whether any test case failed.

Input

```
["PASS", "PASS", "FAIL"]
```

Expected Output

```
true
```

Concept Practiced

- anyMatch()
-

Question 19 — Check if All Tests Passed

Problem

Check whether all test cases passed.

Concept Practiced

- allMatch()
-

Question 20 — Check if No Negative Numbers Exist

Problem

Check whether no negative numbers exist.

Concept Practiced

- noneMatch()
-

Question 21 — Find First Failed Test

Problem

Find first failed test status.

Concept Practiced

- findFirst()
 - Optional
-

Question 22 — Find Any Browser Name

Problem

Find any browser from the list.

Concept Practiced

- findAny()
-

Question 23 — Sum All Numbers

Problem

Find sum of all numbers.

Input

[1,2,3,4]

Expected Output

10

Concept Practiced

- `reduce()`
-

Question 24 — Find Maximum Number

Problem

Find maximum number from the list.

Concept Practiced

- `reduce()`
 - `max()`
-

Question 25 — Find Minimum Number

Problem

Find minimum number from the list.

Concept Practiced

- `min()`
-

Question 26 — Convert List to Set

Problem

Convert list into Set to remove duplicates.

Concept Practiced

- `Collectors.toSet()`
-

Question 27 — Join Browser Names

Problem

Join all browser names using comma.

Expected Output

```
Chrome,Firefox,Edge
```

Concept Practiced

- `Collectors.joining()`
-

Question 28 — Count Total Failed Tests

Problem

Count total failed test cases.

Concept Practiced

- `filter()`
 - `count()`
-

Question 29 — Convert Names to Lowercase

Problem

Convert all names to lowercase.

Concept Practiced

- `map()`
-

Question 30 — Filter Mobile Numbers Starting with 9

Problem

Filter mobile numbers starting with digit 9.

Concept Practiced

- `filter()`
 - `startsWith()`
-

Level 3 — Intermediate (Questions 31–40)

Concepts Covered

- flatMap()
 - groupingBy()
 - partitioningBy()
 - object streams
 - comparator
 - chaining operations
-

Question 31 — Flatten Nested List

Problem

Convert nested list into single list.

Input

```
[[1,2],[3,4]]
```

Expected Output

```
[1,2,3,4]
```

Concept Practiced

- flatMap()
-

Question 32 — Flatten Multiple Test Data Sheets

Problem

Flatten multiple test data sheets into single stream.

Concept Practiced

- flatMap()
-

Question 33 — Group Employees by Department

Problem

Group employees based on department.

Concept Practiced

- `groupBy()`
-

Question 34 — Group Test Cases by Status

Problem

Group test cases into PASS and FAIL.

Concept Practiced

- `groupBy()`
-

Question 35 — Partition Numbers into Even and Odd

Problem

Partition numbers into even and odd groups.

Concept Practiced

- `partitioningBy()`
-

Question 36 — Partition Test Results into Pass and Fail

Problem

Partition test results into pass/fail categories.

Concept Practiced

- `partitioningBy()`
-

Question 37 — Sort Employees by Salary

Problem

Sort employees by salary.

Concept Practiced

- `sorted()`
 - `Comparator`
-

Question 38 — Find Highest Salary Employee

Problem

Find employee with highest salary.

Concept Practiced

- `max()`
 - `Comparator`
-

Question 39 — Extract Employee Names

Problem

Extract only employee names from Employee objects.

Concept Practiced

- `map()`
 - object streams
-

Question 40 — Chain Multiple Stream Operations

Problem

Perform: 1. Filter even numbers 2. Multiply by 10 3. Sort descending 4. Collect into list

Concept Practiced

- stream pipeline
 - chaining
-

Level 4 — Advanced (Questions 41–50)

Concepts Covered

- `parallelStream()`
 - advanced collectors
 - stream optimization
 - real-world SDET scenarios
 - complex object streams
-

Question 41 — Find Duplicate Elements

Problem

Find duplicate numbers from list.

Concept Practiced

- `filter()`
 - Set logic
-

Question 42 — Find Second Highest Number

Problem

Find second highest number using Streams.

Concept Practiced

- sorted()
 - skip()
 - findFirst()
-

Question 43 — Find Longest String

Problem

Find longest browser name.

Concept Practiced

- max()
 - Comparator
-

Question 44 — Count Occurrences of Each Word

Problem

Count frequency of words.

Input

```
["PASS", "FAIL", "PASS"]
```

Expected Output

```
PASS=2  
FAIL=1
```

Concept Practiced

- groupingBy()
 - counting()
-

Question 45 — Validate All API Status Codes are 200

Problem

Check whether all API status codes are 200.

Concept Practiced

- `allMatch()`
-

Question 46 — Find ERROR Logs from Large Log File

Problem

Filter only ERROR logs.

Concept Practiced

- `filter()`
 - file/log processing
-

Question 47 — Process Logs Using Parallel Stream

Problem

Use parallel stream to process large logs.

Concept Practiced

- `parallelStream()`
-

Question 48 — Extract Unique Failed Test Names

Problem

Extract unique failed test names.

Concept Practiced

- distinct()
 - filter()
-

Question 49 — Create Map of Employee Name and Salary

Problem

Convert employee list into Map.

Expected Output

```
{"Amit":50000, "John":70000}
```

Concept Practiced

- Collectors.toMap()
-

Question 50 — Real SDET End-to-End Scenario

Problem

You have a list of test execution results.

Perform below operations:

1. Filter failed tests
2. Remove duplicates
3. Convert names to uppercase
4. Sort alphabetically
5. Collect into final list

Concept Practiced

- Complete stream pipeline
 - Multiple operations together
 - Real-world automation scenario
-

Final Advice for Practice

Do NOT solve questions by memorizing syntax.

Instead:

1. Understand input data
 2. Think what transformation is needed
 3. Think step-by-step pipeline
 4. Build stream gradually
-

Suggested Practice Order

Week 1

Questions 1–15

Week 2

Questions 16–30

Week 3

Questions 31–40

Week 4

Questions 41–50

Most Important Stream Concepts for SDETs

Concept	Importance
filter()	Very High
map()	Very High
collect()	Very High
flatMap()	High
groupingBy()	High

Concept	Importance
reduce()	High
anyMatch()	Very High
Optional	High
Comparator	High
parallelStream()	Medium

Final Learning Goal

After completing these 50 questions, students should be able to:

- process automation test data
- filter API responses
- validate Selenium UI data
- analyze logs
- create reports
- work with collections efficiently
- solve interview problems confidently
- write cleaner automation framework code